Lab Report – 08

Course Code: CSE 412

Course Title: Computer Graphics Lab

Experiment Name: Implementation of 3D Transformation and 3D-object view

Submitted To:

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Submission Date : 10-04-2026

Lab Report: 08

Implementation of 3D Transformation and 3D-object view

Title: 3D Pyramid Translation Using OpenGL

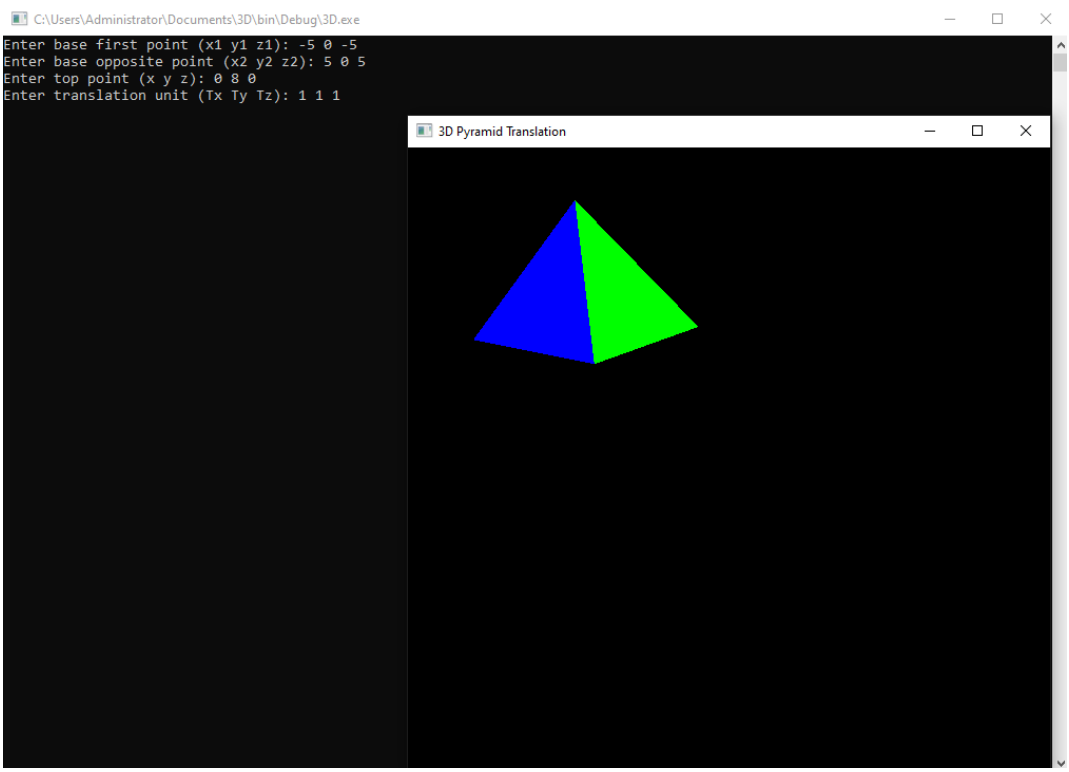
Introduction:

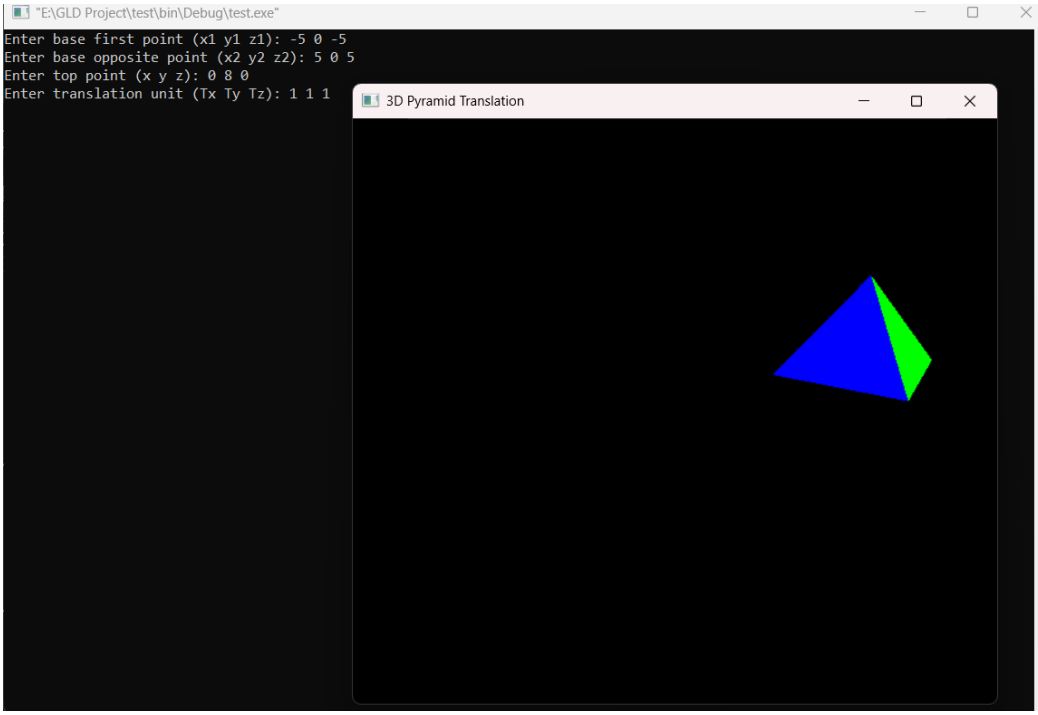
In this project, a 3D pyramid is drawn using OpenGL. The pyramid can be moved in different directions using keyboard input.

Contents:

- **Functions Used:** `drawPyramidFromPoints()`, `glTranslatef()`, `glVertex3f()`, `gluLookAt()`
- **Concept Used:** 3D Translation
- **Shape Used:** Pyramid (base + triangle faces)

Graph:





Draw a 3D pyramid and show movement along X, Y, and Z directions.

Code (Drawing Portion):

```
// ================= KEYBOARD =================
void keyboard(unsigned char key, int x, int y)
{
    switch (key)
    {
    case 'X': moveX += Tx; break;
    case 'x': moveX -= Tx; break;

    case 'Y': moveY += Ty; break;
    case 'y': moveY -= Ty; break;

    case 'Z': moveZ += Tz; break;
    case 'z': moveZ -= Tz; break;
```

```
    }

    glutPostRedisplay();
}

// ================= INIT =================
void init()
{
    glEnable(GL_DEPTH_TEST);

    glMatrixMode(GL_PROJECTION);
    glLoadIdentity();
    gluPerspective(60, 1.0, 1.0, 100.0);

    glMatrixMode(GL_MODELVIEW);
}

// ================= MAIN =================
int main(int argc, char** argv)
{
    cout << "Enter base first point (x1 y1 z1): ";
    cin >> X1 >> Y1 >> Z1;

    cout << "Enter base opposite point (x2 y2 z2): ";
    cin >> X2 >> Y2 >> Z2;

    cout << "Enter top point (x y z): ";
    cin >> TXp >> TYp >> TZp;
```

```
    cout << "Enter translation unit (Tx Ty Tz): ";
    cin >> Tx >> Ty >> Tz;

    glutInit(&argc, argv);
    glutInitDisplayMode(GLUT_DOUBLE | GLUT_RGB | GLUT_DEPTH);
    glutInitWindowSize(600, 600);
    glutCreateWindow("3D Pyramid Translation");

    init();
    glutDisplayFunc(display);
    glutKeyboardFunc(keyboard);

    glutMainLoop();
    return 0;
}
```

Discussion:

The task was to draw and move a 3D pyramid. I used translation to change its position using keyboard keys.

Title: 3D Pyramid Scaling Using OpenGL

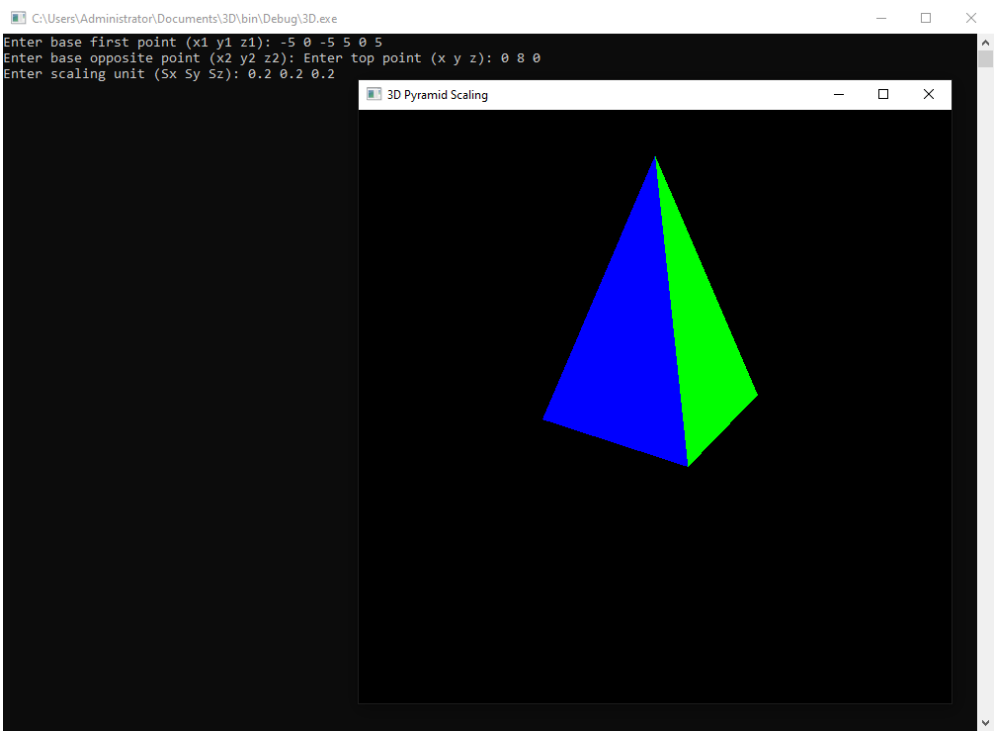
Introduction:

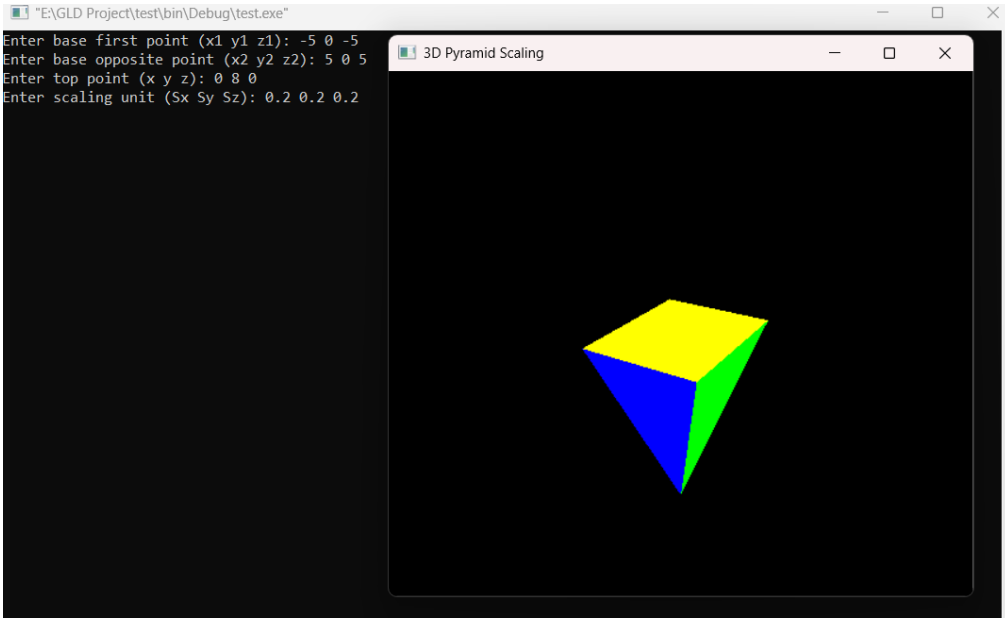
In this project, a 3D pyramid is drawn using OpenGL. The pyramid can be scaled along X, Y, and Z axes using keyboard input.

Contents:

- **Functions Used:** `drawPyramidFromPoints(),glScalef(),glVertex3f(),gluLookAt()`
- **Concept Used:** 3D Scaling
- **Shape Used:** Pyramid (base + triangle faces)

Graph:





Draw a 3D pyramid and show movement along X, Y, and Z directions.

Code (Drawing + Scaling Portion):

```
glLoadIdentity();
gluLookAt(15, 15, 25, 0, 0, 0, 0, 1, 0);

// Apply scaling
glScalef(scaleX, scaleY, scaleZ);

drawPyramidFromPoints(X1, Y1, Z1, X2, Y2, Z2, TXp, TYp, TZp);


// ================= KEYBOARD =================
void keyboard(unsigned char key, int x, int y)
{
```

```
    switch (key)
    {
    case 'X': scaleX += Sx; break;
    case 'x': scaleX -= Sx; break;

    case 'Y': scaleY += Sy; break;
    case 'y': scaleY -= Sy; break;

    case 'Z': scaleZ += Sz; break;
    case 'z': scaleZ -= Sz; break;
    }

    glutPostRedisplay();
}

// ================= INIT =================
void init()
{
    glEnable(GL_DEPTH_TEST);

    glMatrixMode(GL_PROJECTION);
    glLoadIdentity();
    gluPerspective(60, 1.0, 1.0, 100.0);

    glMatrixMode(GL_MODELVIEW);
}
```

```
// ================= MAIN =================
int main(int argc, char** argv)
{
    cout << "Enter base first point (x1 y1 z1): ";
    cin >> X1 >> Y1 >> Z1;

    cout << "Enter base opposite point (x2 y2 z2): ";
    cin >> X2 >> Y2 >> Z2;
    cout << "Enter top point (x y z): ";
    cin >> TXp >> TYp >> TZp;
    cout << "Enter scaling unit (Sx Sy Sz): ";
    cin >> Sx >> Sy >> Sz;

    glutInit(&argc, argv);
    glutInitDisplayMode(GLUT_DOUBLE | GLUT_RGB | GLUT_DEPTH);
    glutInitWindowSize(600, 600);
    glutCreateWindow("3D Pyramid Scaling");

    init();
    glutDisplayFunc(display);
    glutKeyboardFunc(keyboard);

    glutMainLoop();
    return 0;
}
```

Discussion:

The task was to draw a 3D pyramid and allow **interactive scaling** along all axes. Using `glScalef()` the pyramid changes size, and the user can increase or decrease dimensions in real time.

Title: 3D Pyramid Rotation Using OpenGL

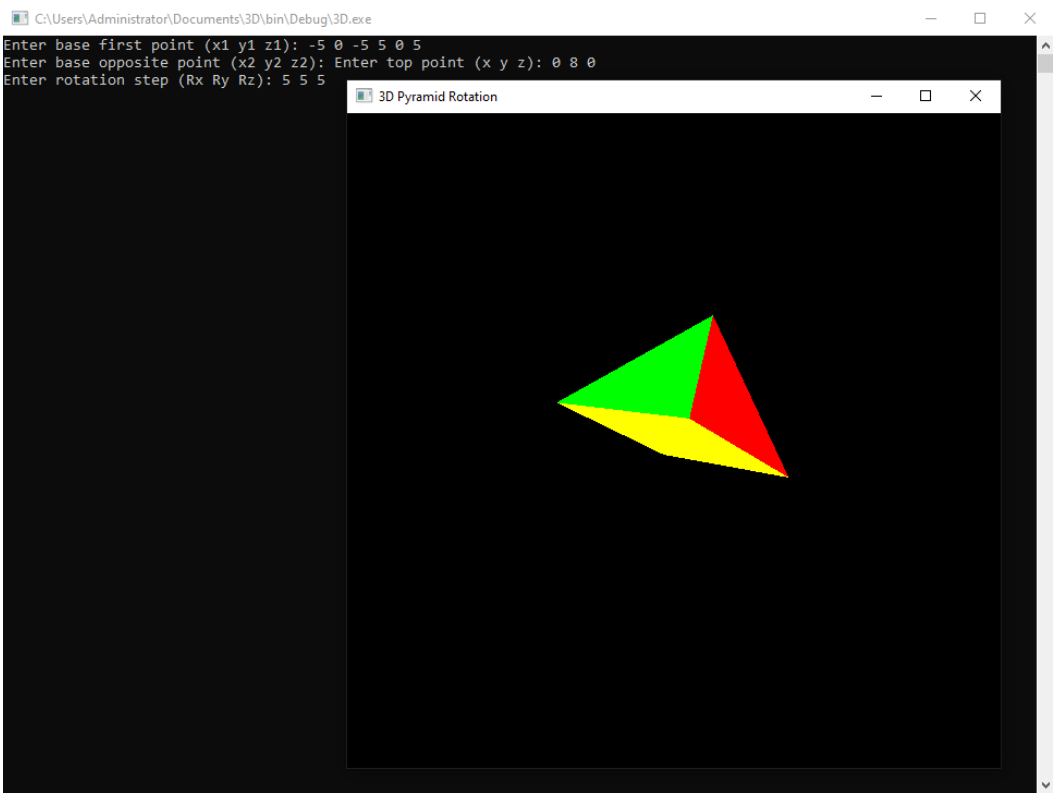
Introduction:

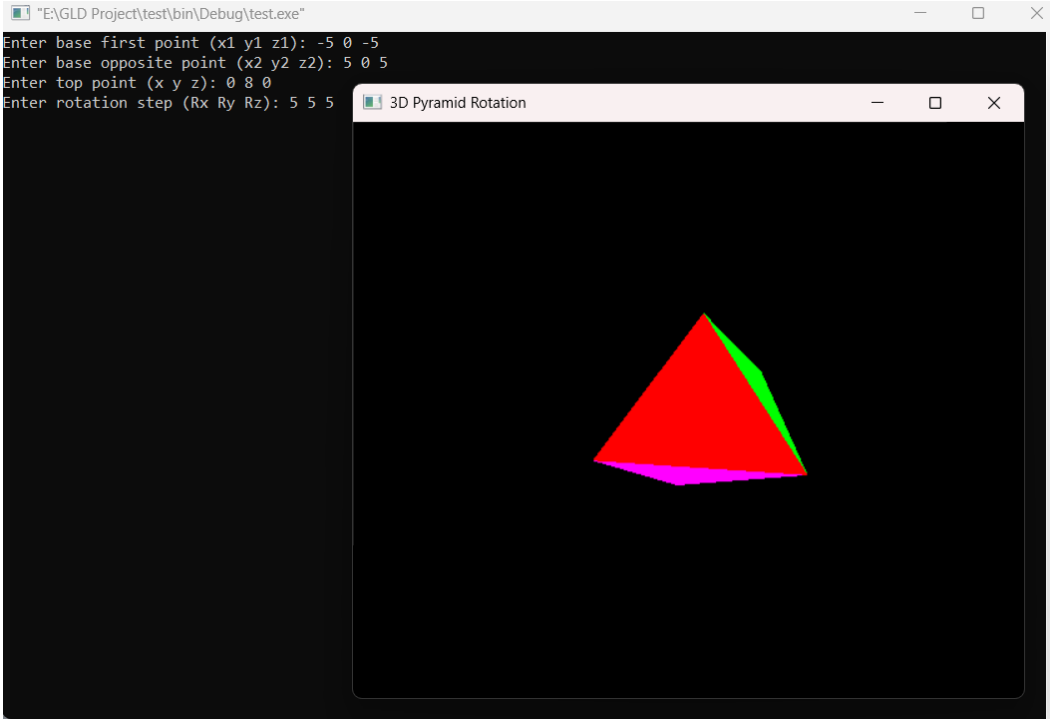
In this project, a 3D pyramid is drawn using OpenGL. The pyramid can be rotated along X, Y, and Z axes using keyboard input.

Contents:

- **Functions Used:** `drawPyramidFromPoints()`, `glRotatef()`, `glVertex3f()`, `gluLookAt()`
- **Concept Used:** 3D Rotation
- **Shape Used:** Pyramid (base + triangle faces)

Graph:





Draw a 3D pyramid and rotate it along X, Y, and Z axes using rotation steps.

Code (Drawing Portion):

```
glRotatef(angleX, 1, 0, 0); // X-axis
glRotatef(angleY, 0, 1, 0); // Y-axis
glRotatef(angleZ, 0, 0, 1); // Z-axis

glBegin(GL_QUADS);
// base
glEnd();
glBegin(GL_TRIANGLES);
// 4 faces
glEnd();
```

```
// ================= KEYBOARD =================
void keyboard(unsigned char key, int x, int y)
{
    switch (key)
    {
    case 'X': angleX += Rx; break;
    case 'x': angleX -= Rx; break;
    case 'Y': angleY += Ry; break;
    case 'y': angleY -= Ry; break;
    case 'Z': angleZ += Rz; break;
    case 'z': angleZ -= Rz; break;
    }
    glutPostRedisplay();
}

// ================= INIT =================
void init()
{
    glEnable(GL_DEPTH_TEST);

    glMatrixMode(GL_PROJECTION);
    glLoadIdentity();
    gluPerspective(60, 1.0, 1.0, 100.0);

    glMatrixMode(GL_MODELVIEW);
}
```

```
// ================= MAIN =================
int main(int argc, char** argv)
{
    cout << "Enter base first point (x1 y1 z1): ";
    cin >> X1 >> Y1 >> Z1;
    cout << "Enter base opposite point (x2 y2 z2): ";
    cin >> X2 >> Y2 >> Z2;
    cout << "Enter top point (x y z): ";
    cin >> TXp >> TYp >> TZp;

    cout << "Enter rotation step (Rx Ry Rz): ";
    cin >> Rx >> Ry >> Rz;

    glutInit(&argc, argv);
    glutInitDisplayMode(GLUT_DOUBLE | GLUT_RGB | GLUT_DEPTH);
    glutInitWindowSize(600, 600);
    glutCreateWindow("3D Pyramid Rotation");

    init();
    glutDisplayFunc(display);
    glutKeyboardFunc(keyboard);

    glutMainLoop();
    return 0;
}
```

Discussion:

The task was to draw and rotate a 3D pyramid. I used rotation to change its orientation along different axes using keyboard keys.